Zhixiao Xiong

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Education Background

University of California, Berkeley Aug 2025 - Now

PhD in Operations Research

Tsinghua University Sep 2021 - Jun 2025

BS in Mathematical Sciences & BEng in Industrial Engineering

Honors and Awards

- First Prize in the 36st China Physics Olympiad Oct 2019
- First Prize in the Tsinghua University Physics Autumn Camp Nov 2019

Areas of Interest

- Stochastic Optimization under Heavy-tailed Noises
- Zeroth-Order Optimization
- Nonsmooth Optimization
- Guided Diffusion Process
- Machine Learning for Optimization

Publications

1. **HEQP: A Hypergraph Neural Network-Based Evolutionary Method for Large-Scale QCQPs.** (An extended journal version of *NeuralQP: A General Hypergraph-based Optimization Framework for Large-scale QCQPs.*) Zhixiao Xiong, Huigen Ye, Hua Xu, and Carlos A. Coello Coelle. *Under review for IEEE Transactions on Cybernetics.*
2. **CurBench: Curriculum Learning Benchmark.** Yuwei Zhou, Zirui Pan, Xin Wang, Hong Chen, Haoyang Li, Yanwen Huang, Zhixiao Xiong, Fangzhou Xiong, Peiyang Xu, and Wenwu Zhu. *ICML 2024: https://proceedings.mlr.press/v235/zhou24o.html* .
3. **NeuralQP: A General Hypergraph-based Optimization Framework for Large-scale QCQPs.** Zhixiao Xiong, Fangyu Zong, Huigen Ye, and Hua Xu. *Arixv preprint: https://arxiv.org/abs/2410.03720v1* .

Research Experience

Function Approximation with ReLU Neural Networks in Two-Stage Stochastic Programming

Independent researcher. Supervised by Prof. Ying Cui, UC Berkeley. Jul 2024 - May 2025

- Improved existing methods by introducing an input-convex neural network to approximate the second-stage objective.
- Proposed a self-supervised training strategy based on the convexity of the neural network to improve data efficiency.
- Analyzed the properties of the second-stage objective to guide the design of ReLU the neural network architecture.

What's Wrong With Graph Neural Networks in Neural Combinatorial Optimization

Research collaborator. Supervised by Prof. Chi-Guhn Lee & Prof. Eldan Cohen, University of Toronto. Mar 2024 - Aug 2024

- Benchmarked existing GNN-based methods for neural combinatorial optimization.
- Proposed a novel training framework for graph combinatorial optimization based on auto-regressive models.
- Conducted part of the experiments and engaged in composing the paper.

A General Hypergraph-based Optimization Framework for Large-scale QCQPs

Research leader. Supervised by Prof. Hua Xu, Tsinghua University. Jun 2023 - Sep 2024

- Proposed a hypergraph-based method to completely encode quadratically constrained quadratic programs as a hypergraph and improved existing hypergraph neural networks to predict optimal solutions.
- Proposed a neighborhood crossover algorithm based on linearization techniques, which enabled small-scale solvers to solve large-scale problems and strengthened traditional large neighborhood search methods by paralleling.
- Led the project with two other undergraduates, developed the code and wrote the paper.
- Proved the equivalence between the process of hypergraph convolution and interior-point methods on quadratic programs.
- (Arxiv preprint) *NeuralQP: A General Hypergraph-based Optimization Framework for Large-scale QCQPs.*

Curriculum Learning Algorithm Development and Open-Source Framework Construction

Research collaborator. Supervised by Prof. Xin Wang, Tsinghua University. Oct 2022 - Oct 2023

- Contributed to the curriculum learning benchmark repo and developed the code based on the RL-teacher paper.
- Conducted experiments for the RL-teacher algorithm by systematically testing all combinations of parameters.
- (ICML 2024 poster) *CurBench: Curriculum Learning Benchmark.*

Selected Course Projects

Repositioning in Free-Floating Bike-Sharing Systems: A Reinforcement Learning Approach

Decision Analytics by Prof. Lei Zhao, Prof. Chen Wang, and Prof. Chengli Zhu, Tsinghua University. Sep 2024 - Dec 2024

- Formulated the problem of intra-cell repositioning in free-floating sharing systems as a Markov Decision Process.
- Proposed a resampling strategy to enhance the data and a data-driven strategy to predict needs using neural networks.
- Developed a free-floating bike-sharing simulation environment for training data generation and model evaluation.
- Currently in progress and will be finished by the end of 2024.

Large Neighborhood Search in Mixed-Integer Linear Programing via Reinforcement Learning

Deep Reinforcement Learning (Graduate) by Prof. Huazhe Xu, Tsinghua University. Mar 2024 - Jun 2024

- Formulated the neighborhood search as a Markov Decision Process and implemented the altorithm with Gurobi and SCIP.
- Developed a training and evaluation framework compatible with multiple RL algorithms and behavior cloning.
- Implemented baseline reinforcement learning algorithms from literature and conducted benchmark experiments.

A Learning Enhanced Decomposition Approach towards the Maximum Clique Problem

Machine Learning for Optimization (Graduate) by Prof. E. B. Khalil, University of Toronto. Sep 2023 - Dec 2023

- Proposed an exact decomposition method to separate graphs for the maximum clique problem via vertex separators.
- Proposed a supervised learning model based on graph features to learn the optimal vertex for decomposition.
- Developed a RL model by representing the identification of vertex separators on graphs as a sequential decision process.

Skills and Others

Language Chinese (native); English (fluent)
Programming Python, Julia, C++, Rust